AeroSentinel

TEAM ASTRO / Design document / Nine-task evaluation environment

Purpose and persistent environment

AeroSentinel evaluates a tool-using agent that diagnoses faults and plans a simplified CubeSat mission. Seeded files, a flight manual, telemetry and mission targets provide observations. A separate verifier owns physical state and success conditions. No policy training is required or claimed.

State persists across actions: time, battery energy, temperature, scalar pointing, radio energy, exposure and delivered data. Tools list/read/write files, run Python, advance orbit and finish. Dispatched ordinary attempts, including tool errors, consume 30 simulated seconds; orbit steps advance 1–30 minutes within a 5,700-second mission. Fixed 30-second integration ticks plus exact event boundaries make equivalent action groupings evolve consistently.

Nine tasks and increasing planning depth

Task group	Decision required
Five guided repairs: EPS, ADCS, thermal, radio and cascade	Diagnose seeded faults and restore safe behavior.
RF energy tradeoff	Balance data rate, link margin and transmitter energy.
Early pass recovery	Repair before a contact deadline that cannot be recovered later.
Observation scheduling	Change payload duty during one useful exposure window.
Science store and forward	Schedule two exposures, drain a finite buffer at two contacts and respect payload/radio energy.

Why the science task requires a sequence

Each window requires at least 90% useful exposure. A constant payload duty therefore needs at least 0.90, while its whole-mission energy allocation permits only 0.101–0.152 across generated durations. Storage holds only 1.05 times the larger window volume, so both windows cannot wait for the final contact. A feasible plan collects in window one, switches off and drains storage, then repeats for window two. Independent feasible alternatives are accepted.

Science is generated only during useful exposure. Contact transmission cannot exceed available buffered data or physical channel capacity. Generated = buffered + delivered + lost. Both window quotas, zero overflow and minimum delivery are mandatory. Writable telemetry and mission files cannot redefine protected truth.

Declared physical assumptions

Solar power uses calibration, cosine pointing and fixed array properties. Battery energy integrates generation minus load. Attitude is scalar drift/damping; temperature uses lumped heat capacity and linear loss. Radio capacity depends on baud, link margin, pointing and contact. The simulator is not flight validated and has no orbital propagator, electrochemistry or three-axis dynamics. Exact equations and active versus descriptive fields are documented in docs/model-assumptions.md.

Verification and evidence

Protected outcomes, bounded execution and source-matched records

Reward and mandatory checks

Standard raw weights are power 0.35, thermal 0.25, pointing 0.20 and communications 0.20. Observation scales these by 0.8 and adds exposure 0.20. Science scales them by 0.6 and adds window completion 0.20 and actual science delivery 0.20. Progress is clamp((raw – initial no-op raw)/(1 – initial no-op raw), 0, 1). Every mandatory check must pass for reward at least 0.95. Invalid fields score zero; catastrophic failures remain latched. Final scoring projects remaining time without rewriting accumulated history.

Isolation and common model contract

Verifier code and mission truth remain in the harness. Agent workers use sandbox UID 1001; behavioral fixtures use UID 1002. Authenticated controls, read-only root, internal networking, resource limits and descendant cleanup bound execution. Docker and the host kernel remain trusted. RF fixtures include the live configuration and full-schema variants, preventing fixture-only correctness claims.

The runner allows 40 replies, 10 executed tools per reply and 600 seconds. Tool operations have 60 seconds and final verification a separate 120 seconds. Reference/no-op baselines share 120 seconds per task and gate API use. Cancellable provider calls have at most two transient retries. Malformed received replies count as steps. Inputs, raw replies, actions, results, errors, hashes, rewards, tokens and timings are logged. Large responses are bounded before buffering; history removes complete signed exchanges when necessary. MODEL/API_KEY/BASE_URL and exact append text use one interface.

Evidence on this source and expected build contract

Check	Observed result
Linux regressions / seeded sweep	150 tests passed; 180 task/seed pairs across all nine tasks.
Container isolation / all-task gates	12 tests passed; all nine reference = 1.0 and no-op = 0.0.
Scripted policies, 45 cases each	No-op 0; nominal 25; max-radio 20; privileged reference 45 successes. Both shortcut policies fail all 20 hard cases.
Held-out mission stress	474 expectations pass, including independent feasible plans and correctly rejected near-boundary decisions. 450 simulator + 24 private HTTP cases.
Synthetic client conformance	13 tests; 32 loopback HTTP requests. Exact append, malformed recovery, context limits and cancellation; zero model calls.
Six deliberate failure probes	Relaxed targets, forged telemetry, missed contact/exposure, science overflow and forged science inventory all fail.
Four organizer-model attempts	3 successes in 4 attempts on gpt-6-luna. Per-task outcomes and provider errors are preserved in validation-evidence.json.

Scripted references and attack probes do not measure model reasoning. Local synthetic provider tests do not prove external service reliability. Zero-reply attempts are diagnostic and do not count toward three completed model-run logs. Times use the harness monotonic clock; host/VM pauses limit independent external wall-clock claims. Provenance v3 binds source and six expected build inputs, not a measured Docker daemon. Historical results are not relabeled.

Reproduction and submission status

One Compose command builds and runs the selected task. CI checks an isolated empty-cache build against 180 seconds. Packaging binds extra validation-code and HTTP-fixture hashes, checks case counts, reviewed PDFs, checksums and secret scans. Astro.zip is the reference archive. Public repository visibility is user requested and conflicts with the PS private-repository rule. Exact Astro/astro case, organizer access, deadline eligibility and form upload remain unconfirmed.

Evidence: submission_bundle/validation-evidence.json. Harness b75030d6ecd33e8f80… Sandbox 2e0b2933cd17ebf44f… Expected build 8be65dbc656d82abe0…